

Program: Diploma in Robotic Process Automation	
Course Code: 3302	Course Title: Digital Design and Computer Architecture
Semester: 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To introduce the concept of Digital and Binary Systems.
- To provide basic knowledge of Combinational logic circuits.
- To provide basic knowledge of Sequential logic circuits.
- Understand the basic concepts, structure and operation of processor and different types of memories.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge in Electronics	2031	Fundamentals of Electrical & Electronics Engineering	II

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	To understand Digital Systems, data representation, to know Logic Gates and Boolean Algebra and to understand Gate Level Minimization	10	Applying
CO2	To understand combinational logic circuits	11	Applying
CO3	To understand the working of flip-flops, shift registers and working of various counters	14	Understanding

CO4	Describe the fundamental units of a computer. Explain the basic concepts of memory and categorize different types of memories	10	Understanding
	Series Test		

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	2				
CO2	3	3	3				
CO3	3	3	3				
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To understand Digital Systems, data representation, to know Logic Gates and boolean Algebra and to understand Gate Level Minimization		
M1.01	To understand various number systems To state the need of a binary number system in modern digital technology To explain the conversion from decimal to binary and vice versa with suitable examples To list the features of Hexadecimal number system with suitable examples To explain the Conversion of hexadecimal into decimal and binary and vice versa To explain binary arithmetic such as addition, subtraction, multiplication and division with examples To explain binary subtraction using 1's complement and 2's complement method	3	Understanding

M1.02	Simplification of Boolean expressions To state Demorgan's theorems To state the rules and laws of Boolean algebra To explain Sum Of Product (SOP) expression, Product Of Sum (POS) expression, minterms and maxterms To state the need for simplifying Boolean expression To simplify Boolean expressions with the help of logic rules and truth tables. To state the basic principle of Karnaugh map 1.2.7 To explain two variables, three variables and four variables K-maps and their reductions with the help of suitable examples To state Don't care terms and their role in solving K-maps To list the advantages and disadvantages of K-map	4	Applying
M1.03	To know Logic Gates To explain the operation of AND, OR, NOT, NAND, NOR, EXOR and EXNOR with their symbols and truth tables To realize AND, OR, NOT, EXOR and EXNOR using universal gates	3	Understanding
Contents: - Number systems and logic gates Number systems - decimal, binary, octal and hexadecimal number systems - conversion - binary arithmetic. Boolean Algebra – Introduction- Basic definitions – Axiomatic Definition of Boolean Algebra - Basic Theorems and Properties of Boolean Algebra – Boolean Functions- Canonical and standard forms — Minimization of Boolean function using K-Map (up to 4 variables) Digital Logic Gates –Introduction to logic gates - universal property of NAND and NOR gates - realization of Basic gates			
CO2	To understand combinational logic circuits		
M2.01	To understand the combinational logic circuits To describe combinational logic circuits To design half adder, full adder, half subtractor, and full subtractor To explain BCD adder and parallel adder	4	Understanding
M2.02	To explain Binary Multiplier and Magnitude Comparators	2	Understanding

M2.03	To explain the operation of 4x1 Multiplexer and 1x4 De-multiplexer To list the various applications of Multiplexers and De-multiplexers	3	Understanding
M2.04	To explain the operation of 3bit encoder and decoder.	2	Understanding
	Series Test – I	1	
Contents : Combinational Circuits – Binary Adder –Subtractor- Parallel Adder- BCD Adder- Binary Multiplier – Magnitude Comparators-Decoder –Encoder-Multiplexer - Demultiplexers.			
CO3	To understand the working of flip-flops shift registers and working of various counters		
M3.01	To understand the working of flip-flops To describe sequential logic circuits To distinguish between synchronous and asynchronous sequential logic circuits To explain latches & flip-flops To explain SR & JK flip flop using NAND & NOR gates To state the race around condition in JK flip flop and list the methods for eliminating race around condition To explain the working of master slave JK flip flop (block level explanation only) To explain D and T flip flops with their characteristic tables	5	Understanding
M3.02	To understand shift registers To explain the functions of Shift registers To explain the working of shift registers:- serial-in serial-out, parallel-in parallel-out, parallel-in serial-out and serial- in parallel-out To differentiate between right shift and left shift registers To explain the working of ring counter and its applications To explain the working of Johnson counter and its applications	5	Applying
M3.03	To understand the working of Counters To differentiate between synchronous and asynchronous counters To implement mod-10 asynchronous counter using JK flip flop To explain mod-8 synchronous counter and its realization using JK flip flop To explain 3 bit up-down counter using JK flip flop	4	Applying

Contents Sequential Circuits – Storage elements – Latches & Flip-Flops Registers – Shift register – Ripple Counters- Synchronous Counters-Ring counters - Johnson Counter			
CO4	Describe the fundamental units of a computer. Explain the basic concepts of memory and categorize different types of memories		
M4.01	To understand the Functional Units of Computers: Input Unit, Memory Unit, Arithmetic and Logic Unit, Output Unit, Control Unit.	2	Understanding
M4.02	To understand the Processor Architecture – General internal architecture, Address bus, Data bus, control bus. Register set – status register, Accumulator, program counter, stack pointer, general purpose registers.	3	Understanding
M4.03	To understand The Memory System: Basic Concepts, Semiconductor RAM Memories, Internal Organization of Memory Chips. To differentiate between Static Memories, Dynamic RAMs. Read-only Memories.	3	Understanding
M4.04	To differentiate between Cache Memories and Virtual memory. To explain Direct Memory Access, Memory Hierarchy	2	Understanding
	Series Test – II		
Contents: Functional units of computer – Input Unit-Memory Unit -Arithmetic and Logic Unit - Output Unit -Control Unit. Processor architecture3, Memory System - Semiconductor RAM Memories. Internal Organization of Memory Chips - Static Memories- Dynamic RAMs- Read-only Memories. Cache Memories and Virtual memory. DMA, Memory Hierarchy.			

Text / Reference

T/R	Book Title/Author
T1	1. A Anand Kumar- Fundamentals of digital circuits PHI
T2	Floyd and Jain -Digital Fundamentals - Pearson- 8th Edition
R1	Malvino and Leach -Digital Principles and Applications - Tata McGraw-Hill
T3	Computer Organization By Carl Hamacher, Fifth Edition, Mc Graw Hill
T4	Computer Architecture and Organisation By John P Hayes, Mc Graw Hill

Online Resources

Sl.No	Website Link
1	http://nptel.ac.in/course :s/ Webcourse- contents/IIT-%20Guwahati/comp_org_arc/web/
2	http://nptel.ac.in/course :s/ Webcourse- contents/IIT-Kharagpur